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The Chota basin and its significance for the inception and tectonic setting of the inter-Andean depression in Ecuador

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Abstract

The inter-Andean structure is defined as an approximately north–south-trending, linear, topographic depression in Ecuador located between the Cordillera Real and the Cordillera Occidental. The depression swings westward toward the Gulf of Guayaquil in southern Ecuador, dissecting the topography of the Cordillera Occidental. The structural limits of the depression are reactivated faults, which formed during accretionary events after 140 Ma along the Ecuadorian continental margin. Several distinct basins progressively formed along the depression. A new, radiometric-based, chronostratigraphic framework for the sedimentary series of the Chota basin has been combined with data from other subbasins in the inter-Andean depression to reevaluate the timing and formation of the larger scale tectonic structures. Inception of the Chota basin commenced at ∼6–5 Ma (latest Miocene), and the progressively younger Quito-San Antonio-Guayllabamba, Ambato-Latacunga, and Riobamba-Alausí basins formed along a southward trend. Each basin was filled with alluvial fan, fluvial, lacustrine, and contemporaneous volcanic deposits. Synsedimentary transpressive deformation was dominant during basin development, though minor, synsedimentary, normal faulting is assumed to have occurred during short periods of weakening compression. The inter-Andean depression of Ecuador formed in the vicinity of a major restraining bend, which accommodates the northward displacement of the north Andean block with respect to the South American plate. A ramp basin model is proposed to explain the tectonosedimentary development of the inter-Andean depression.

Keywords: Chota basin; Ecuador; Inter-Andean depression; Pliocene–Pleistocene; Stratigraphy; Tectonics

Resumen

La estructura de la Depresión Interandina está definida como una depresión topográfica de dirección norte-sur entre la Cordillera Real y la Cordillera Occidental de Ecuador. En el sur, la depresión se desvía hacia el oeste en dirección del Golfo de Guayaquil, disectando la topografía de la Cordillera Occidental. Los límites estructurales de la depresión corresponde a fallas reactivadas, las cuales fueron formadas anteriormente durante eventos acrecionarios a lo largo del margen continental ecuatoriano. Durante el desarrollo de la Depresión Interandina varias cuencas se formaron progresivamente. Un nuevo marco cronoestratigráfico basado en edades radiométricas de las series sedimentarias de la Cuenca de Chota ha sido combinado con datos de otras subcuencas en la Depresión Interandina con el propósito de reevaluar la edad y formación de la estructura tectónica a mayor escala. La formación individual de las cuencas de menor escala se propagó desde 6–5 Ma (Mioceno tardío) desde la Cuenca de Chota ubicada al norte hacia el sur en las cuencas de Quito-Guayllabamba, Ambato-Latacunga y Riobamba-Alausí. Las diferentes subcuencas fueron llenadas con sedimentos de abanico aluvial, fluviales, lacustres y depósitos volcánicos contemporáneos. Deformación transpresiva syn–sedimentaria prevaleció durante gran parte del desarrollo de las cuencas. Se asume que fallas normales syn–sedimentarias menores han ocurrido durante cortos períodos de moderada compresión. La Depresión Interandina de Ecuador se formó en la vecindad de una

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estructura mayor de tipo 'restraining bend,' la cual acomoda el desplazamiento hacia el norte del bloque andino con respecto a la placa Sudamericana. Un modelo de cuenca de rampa es propuesto para explicar el desarrollo tectono-sedimentario de la Depresión Interandina.

1. Introduction

The inter-Andean depression (IAD) is an approximately N–S- to NNE–SSW-oriented topographic depression spanning between ~2°30′S and ~0°45′N in Ecuador. The IAD is flanked by the Cordillera Real (CR) to the east and the Cordillera Occidental (CO) to the west (Fig. 1) and hosts a series of sedimentary basins (hereafter referred to as subbasins) that formed in response to large-scale, late Miocene-Recent tectonic rearrangements in the Ecuadorian Andean forearc and arc. Traversing from north to south, the following subbasins have been recognized in Ecuador: (1) the Chota basin, located in the northern IAD between the towns of Ibarra and Tulcán (Fig. 1); (2) the Quito-San Antonio-Guayllabamba basin; (3) the Ambato-Latacunga basin; and (4) the Riobamba–Alausí basin in the extreme south.

Previous studies have assumed that these basins were active since the Neogene (e.g. Winter and Lavenu, 1989a,b; Egüez and Beate, 1992; Lavenu et al., 1995, 1996; Baragán et al., 1996; Ego and Sébrier, 1996) and used chronostratigraphy, sedimentology, structural analysis, and seismicity to propose several different mechanisms for the formation of the IAD. Early works invoke a simple graben structure that formed during E–W-oriented extension (e.g. Baldock, 1982). Later studies recognize compressional pulses, which resulted in the synsedimentary inversion of the subbasins (e.g. Winter and Lavenu, 1989b; Lavenu et al., 1992; Ego and Sébrier, 1996) and which they relate to transpressional movements driven by the dextral displacement of the CO with respect to the CR. Ego et al. (1996) add to this transpressional model and propose that the IAD formed along a restraining bend in a dextral transpressive system involving the Pallatanga-Pujilí-Calacalí fault in the south and the Chingual-La Sofia fault in the north (Fig. 1). However, Tibaldi and Ferrari (1992) propose a kinematic model based on sinistral wrenching (associated with piggy-back basin formation), because the CR was displaced approximately northward at a greater rate than was the CO. More specifically, Baragán et al. (1996) propose that the Chota basin initially formed in a W–E to WNW–ESE extensional regime and then was compressed in a similar orientation. Winkler et al. (2002) suggest that the IAD represents a spindle-shaped basin structure that opened and closed in scissor-like movements between the two cordilleras.

A useful attempt at reconstructing the processes that have driven basin formation and deformation in the subbasins of the IAD and the IAD itself requires the following: (1) a clear definition of the morphology, extent, and boundaries of the IAD; (2) a well-constrained sedimentological and chronostratigraphic framework; and (3) knowledge of the nature, duration, and driving forces of structural deformation. Many local and regional field observations from the sedimentary rocks of IAD subbasins have been published. However, most tectonic interpretations—particularly those from the Quito, Guayllabamba, and Chota areas—lack an appropriately constrained chronostratigraphic framework. For example, no reliable chronostratigraphic framework has been published for the Chota basin. We present a synthesis of recent work (Abegglen, 2001; Tobler, 2001; Villagómez et al., 2002; Villagómez, 2003) that includes detailed radiometric, heavy mineral, and field data from the Chota, Guayllabamba, and Quito basins. In particular, we determine a detailed chronostratigraphic framework for the strata of the Chota basin using apatite and zircon fission track (AFT and ZFT, respectively) analyses, which, in combination with evolutionary models for the bounding cordilleras, significantly reinterpret the extent of the IAD, the timing of its inception, and the geodynamic context. These data also clarify that the IAD has a different history than the intermontane basins in southern Ecuador (e.g. Cuenca, Loja; Hungerbühler et al., 2002).

2. Definition of the Plio-Pleistocene IAD in Ecuador

In Ecuador, the IAD geographically extends from ~2°30′S to the Colombian border and is characterized by a row of depressions below 3000 m between the CR and CO (Fig. 1). The main structural limits of the regional depression are reactivated crustal scale faults, which formed during successive Cretaceous and early Tertiary accretionary events along the Ecuadorian continental margin (Litherland et al., 1994; Spikings et al., 2001; Hughes and Pilatasig, 2002). These faults can be traced along the northern Andean chain in Colombia (Toussaint and Restrepo, 1994), but their origin remains controversial. The possible continuation of the IAD structure into Colombia (Cauca-Patia depression) is not discussed here.

The Peltetec fault defines the eastern limit of the IAD and may represent a late Jurassic structure that formed during the accretion of the terranes that constitute the CR (Litherland et al., 1994). Alternatively, the Peltetec fault may have formed during the Late Cretaceous accretion of the oceanic Pallatanga terrane (Spikings et al., 2005). The Pallatanga–Pujilí–Calacalí fault, which formed during the late Cretaceous accretion of the Pallatanga terrane, defines the western border of the IAD (Fig. 1). Parallel with the Pallatanga–Pujilí–Calacalí fault, the IAD swings westward south of ~S2°10′, toward the Gulf of Guayaquil, and dissects the topography of the CO (Fig. 1; Lavenu et al., 1996).

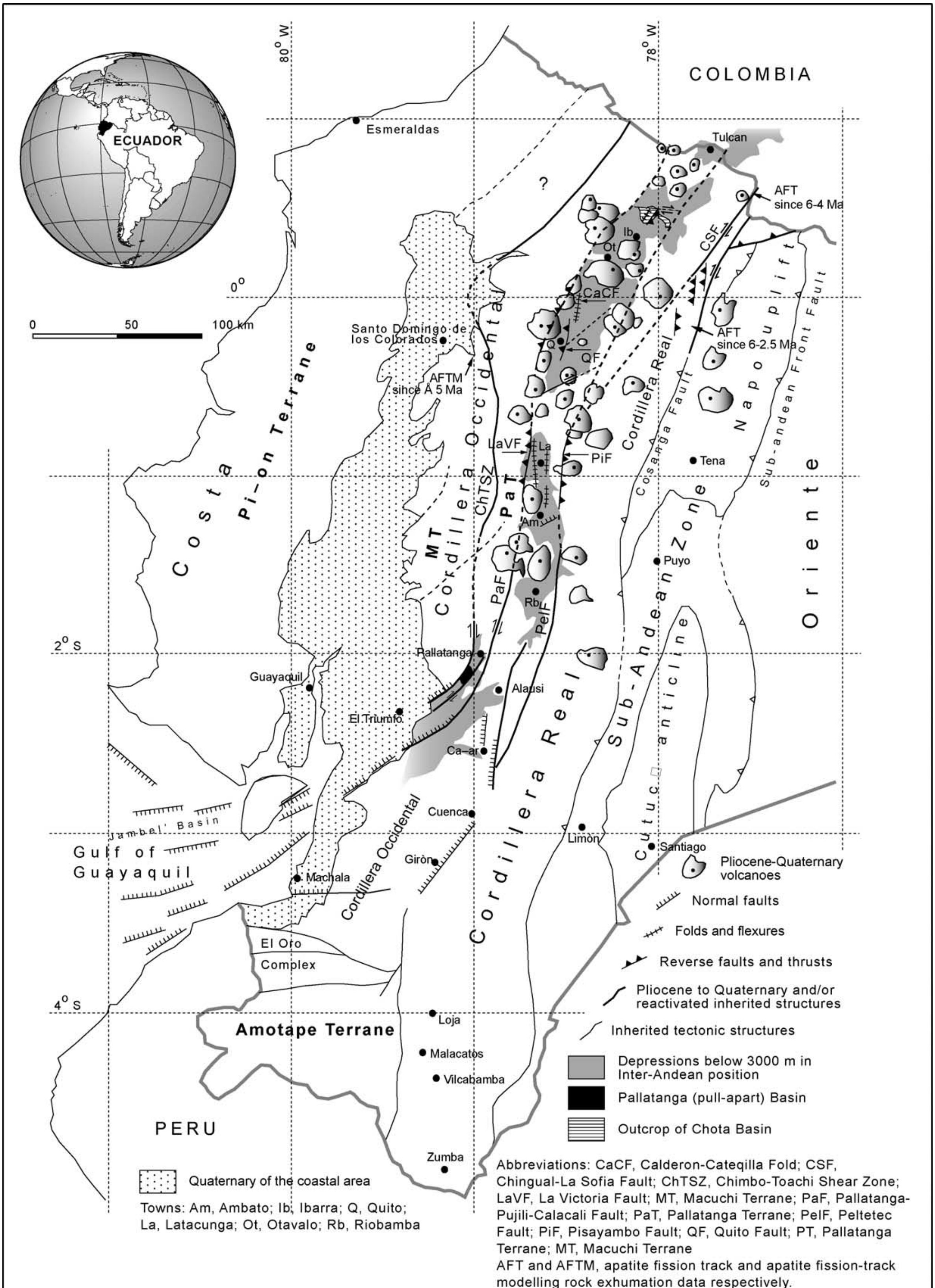

Fig. 1. Tectonic sketch of the IAD structure based on Winter and Lavenu (1989a,b), Tibaldi and Ferrari (1992), Litherland et al. (1993, 1994), Lavenu et al. (1995), Ego and Sébrier (1996), Ego et al. (1996), and our own observations. The IAD structure occurs as a topographic depression between the CR to the east and the CO to the west. The trend of the structure is depicted by the gray shading. The depression swings westward toward the Gulf of Guayaquil south of 2°S, cross-cutting the topography of the CO.

This area also is characterized by the opening of the Pallatanga pull-apart basin, which has been forming since ~2.5 Ma (Winter and Lavenu, 1989a) along a short stretch of releasing bend. General dextral transpressive movement in the forearc and arc is compensated for by extension in the Jambeli basin in the Gulf of Guayaquil (e.g. Benítez, 1995; Hungerbühler et al., 2002).

The sedimentary sequences of the Chota, Quito-San Antonio-Guayllabamba, Ambato-Latacunga, and Riobamba-Alausí subbasins range between latest Miocene and Pleistocene. As noted by Lavenu et al. (1996), the stratigraphic record in the IAD implies that the structural evolution of the IAD was different than that of the middle and late Miocene arc and forearc basins in southern Ecuador (Cuenca, Girón-Santa Isabel, Nabón, Loja, Catamayo–Gonzanamá, Malacatos–Vilcabamba; e.g. Hungerbühler et al., 2002). Volcanic activity in the Ecuadorian Andes since ~5 Ma (Pliocene–Quaternary) has been restricted to the region north of the town of Pallatanga and concentrated along the bordering faults of the IAD (e.g. Barberi et al., 1988). The concentration of volcanic activity has been temporally and geographically coeval with the inception and tectonic development of the IAD.

Miocene and younger reactivation and exhumation of the deformed rocks exposed in and proximal to the main faults in the cordilleras that bound the IAD has been constrained by AFT analysis (Fig. 1). Sheared rocks from the Chimbo-Toachi shear zone (CO) were rapidly cooled and exhumed at ~5 Ma (Spikings et al., 2001, 2005). In addition, basement Jurassic volcanic rocks (Misahualli Formation), located between the Chingual-La Sofia and Cosanga faults in the northernmost CR, have been cooling rapidly since 6–4 Ma (Spikings et al., 2000). Finally, tectonically sliced Jurassic volcanoclastic rocks located at approximately 0°45′S in the sub-Andean zone (Paradalarga unit; Vallejo and Buitron, 1999) were cooled and exhumed during ~6–2.5 Ma (Ruiz, 2002). We attribute these exhumation events to Late Miocene and younger, large-scale fault reactivation during the formation of the IAD.

3. Geological framework of the subbasins of the IAD

The sedimentary sequences of the IAD overlie either exposures of basement rocks of the cordilleras (Pallatanga and Guamote units; Litherland et al., 1994; Hughes and Pilatasig, 2002; Villagómez et al., 2002) or Oligocene–late Miocene volcanic successions, which are also exposed to the south of the present IAD in the sierra of southern Ecuador (Fig. 2). Chronostratigraphic and lithologic correlations suggest that the locally defined volcanic Huigra Formation correlates with parts of the Saraguro group in southern Ecuador. Similarly, the Alausí and Pisayambo Formations are equivalent to the Turi and Tarqui Formations in southern Ecuador (Hungerbühler et al., 2002).

3.1. Chota basin

The ZFT and AFT analyses (Table 1, Fig. 3) were performed on various sedimentary and volcanic rocks from the Chota basin using the methodology described by Spikings et al. (2001). Generally, zircons revealed smaller $\pm 1\sigma$ errors than did apatites. When the single grain fission track ages for an individual sample yield a $p(\chi^2)$ value of $<5\%$, more than one age population is assumed to be present, and individual age populations have been resolved (Galbraith and Green, 1990). If two or more age populations occur in the same volcanic rock, the youngest age is considered the eruption or stratigraphic age.

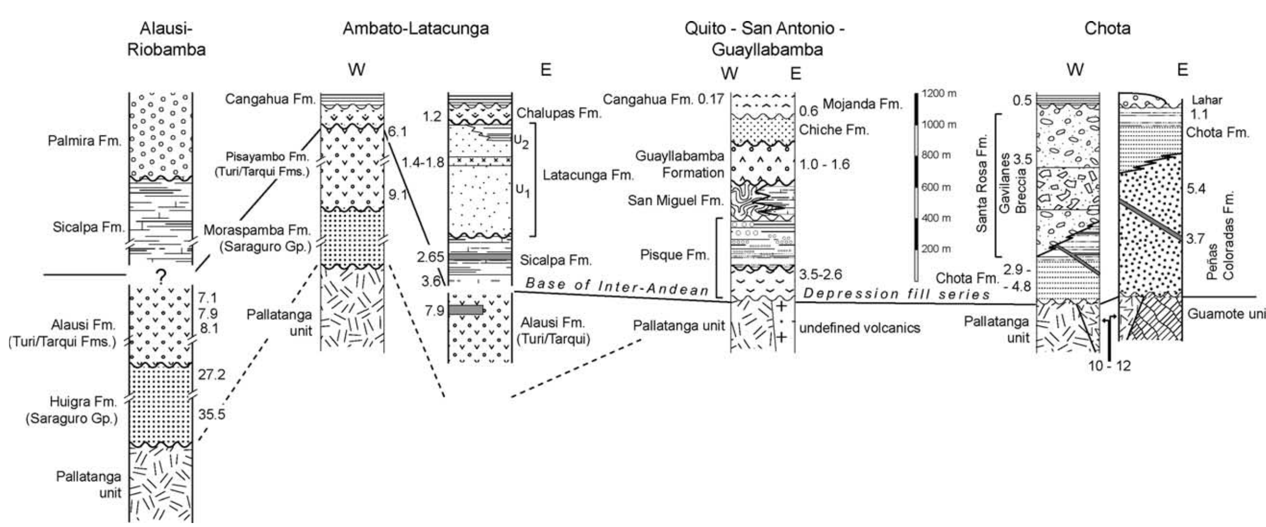

Fig. 2. Composite lithologic profiles of the main IAD subbasins. Numbers indicate chronostratigraphic ages and age ranges in Ma. Chota basin numbers are ZFT and AFT ages (Fig. 3, Table 1); others are K/Ar and $^{40}Ar/^{39}Ar$ ages. Compiled from Barberi et al. (1988), Egüez et al. (1992), Egüez and Beate (1992), Lavenu et al. (1995, 1996), Abegglen (2001), Tobler (2001) and Villagómez (2003).

Table 1

Apatite and zircon fission track data from the Chota basin, northern IAD, Ecuador (sample location in UTM grid reference)

Sample number	No. of grains	Standard track density $\times 10^6$	Spontaneous track density $\times 10^6$	Induced track density $\times 10^6$	P (c^2) %	U (ppm)	FT age ± 1 s (Ma)	Mean track length (mm)	SD	Formation	Lithology	UTM
Apatite												
AT2	30	1.087 (5920)	0.00577 (4)	0.4170 (289)	59	5	2.9 $\pm$ 1.5			Chota	Ash (30 cm thick)	826975/ 51350
AT3	30	1.103 (5920)	0.00726 (13)	0.3782 (677)	83	4	4.1 $\pm$ 1.2			Chota	Ash (8 cm) 5 m below main gas-tropod horizon	826325/ 51125
AT5	30	1.119 (5920)	0.05230 (5)	0.3033 (290)	99	3	3.7 $\pm$ 1.7	15.54 $\pm$ 0.00 (1)	0	Peñas Coloradas	Andesitic dyke (4 m thick)	830800/ 51725
AT13	30	1.233 (5920)	0.001523 (3)	0.3416 (673)	48	4	1.1 $\pm$ 0.6			Chota 10 m below lahar	Ash (25 cm thick)	828875/ 49860
AT15	21	1.249 (5920)	0.01501 (11)	0.3110 (228)	97	3	12 $\pm$ 4			Volcanic base-ment block	Ash (10 cm thick)	828925/ 49940
AT17	29	1.265 (5920)	0.01162 (26)	0.2712 (607)	98	3	11 $\pm$ 2	14.09 $\pm$ 0.2 (16)	1	Volcanic base-ment	Andesite (5 m) between Ambu-qui Gr. and Chota Fm. ~	832150/ 49450
AT23	32	1.329 (6103)	0.01027 (37)	0.2526 (910)	100	2	10 $\pm$ 2	15.64 $\pm$ 0.00 (1)	0	Volcanic base-ment block	Ash (40 cm thick)	830200/ 51675
Zircon												
AT7	6	0.4131 (2590)	4.217 (355)	4.098 (345)	73	397	28 $\pm$ 2.0			Chota	Ash (25 cm thick)	826925/ 42475
	5	0.4131 (2590)	1.317 (139)	3.828 (404)	45	371	9.6 $\pm$ 1.0					
	17	0.4131 (2590)	0.4996 (135)	3.205 (866)	80	310	4.3 $\pm$ 0.4					
AT4	24	0.3762 (2590)	5.238 (1623)	3.753 (1163)	0	399	35 $\pm$ 2			Peñas Coloradas	Fine volcaniclas-tic breccia	830125/ 49850
	14	0.3762 (2590)	2.333 (609)	10.65 (2781)	0	1133	5.4 $\pm$ 0.4					
AT3	20	0.3331 (2590)	0.4042 (250)	1.875 (1160)	41	225	4.8 $\pm$ 0.4			Chota	Ash 5 m below gastr. horiz.	826325/ 51125
AT6	20	0.3454 (2590)	0.04163 (8)	1.905 (366)	9	221	0.5 $\pm$ 0.2			Unconformable cover series (poss. corr. with Chiche Fm.)	White unconsoli-dated ash (50 cm thick)	827150/ 52325
AT10	19	0.3577 (2590)	0.4564 (2590)	3.161 (1697)	70	354	3.5 $\pm$ 0.3			Santa Rosa (above Gavilanes breccia)	Ash (50 cm)	825625/ 52450
AT11	20	0.3638 (2590)	0.7655 (593)	5.219 (4043)	0	574	3.2 $\pm$ 0.3			Chota (middle-upper)	Ash (40 cm)	828175/ 51900

Numbers in parentheses are the number of tracks counted; no track lengths are measured in the zircon crystals. When $p(c^2)$ is $<5\%$, the fission track age is the *central* age; otherwise, it is the *pooled* age. Dating was carried out by Richard Spikings for Peter Abegglen and Stefan Tobler (all ETH-Zurich). Note that in sample AT3, apatites and zircons are dated.

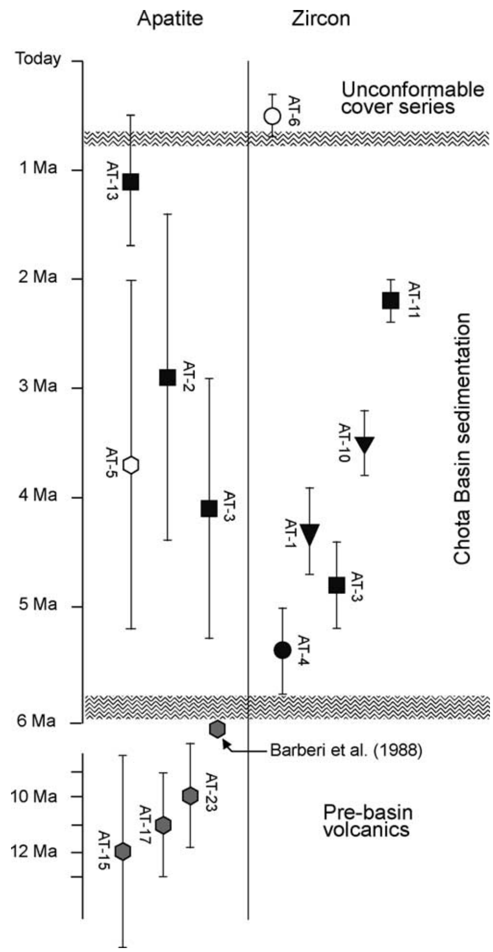

Fig. 3. Graphical presentation of AFT and ZFT ages ($\pm 1\sigma$) derived from the Chota basin. Squares are the Chota Formation; triangles are the Santa Rosa Formation; filled circles are the Peñas Coloradas Formation; filled polygons are prebasinal volcanics; open polygons are folded dykes; and open circles are unconformable cover series. See Table 1.

We present heavy mineral data from the Chota basin in Fig. 4, and the interpretation of each mineral assemblage follows the general rules presented by Mange and Maurer (1991).

The sedimentary sequence in the Chota basin is ~1200–1400 m thick and geographically divided into two parts by an extensive N–S-flowing lahar sheet, across which (W–E) lithologic correlations are difficult (Fig. 5). Sedimentary rocks unconformably overlie andesites and volcanic ashes dated at 12±4, 11±2, and 10±2 Ma (AFT; Table 1, samples 15, 17, 23; Fig. 3) in the eastern and central region. The volcanic rocks unconformably overlie low- to medium-grade metamorphosed and highly deformed rocks (slates and quartzites) of the Guamote terrane (Ambuquí group), which is mainly exposed in the CR (Litherland et al., 1994). No basement-cover contact is observed in the western basin, though the close proximity of pillow basalts of the Pallatanga unit of the CO suggests that the sedimentary rocks may partly overlie this unit. Volcanic rocks also may lie stratigraphically between the basement and the sedimentary basin fill in the west. Barberi et al. (1988) report K/Ar ages of 6.3±0.03 and 6.31±0.1 Ma (Fig. 3) for andesites in the west, which underlie the sedimentary rocks of the Chota basin, though they do not provide sample locations.

A fine matrix of volcanoclastic breccia from the Peñas Coloradas Formation yields a ZFT age of 5.4±0.4 Ma, and the sequence is cut by a dyke that yields an AFT age of 3.7±1.7 Ma (Table 1, Figs. 2 and 3). The ZFT age is interpreted as an eruptive age, which, in contrast with previous studies, indicates that the Peñas Coloradas is the same age or may predate the Chota Formation. Previous K/Ar ages of ~6.3 Ma (Barberi et al., 1988) from the volcanic basement of the Chota basin corroborate our postulated date for the inception of the sedimentary basin during the latest Miocene.

Both the heavy mineral content and paleoflow analysis of alluvial fans in the Peñas Coloradas Formation indicate that it was derived from the east (Baragán et al., 1996; Abegglen, 2001; Tobler, 2001). The heavy mineral assemblage is diagnostic of a source terrane composed of medium- to high-grade, regional metamorphic rocks (Fig. 4; garnet, epidote, clinozoisite, zoisite, kyanite) and granitoids (zircon, tourmaline, rutile), such as those that crop out in the CR. The Peñas Coloradas overthrusts the Chota Formation along a steep, west-verging fault proximal to the east of the lahar (Egüez and Beate, 1992; Abegglen, 2001; Fig. 5).

There is no mapable evidence of the stratigraphic relationship between the Chota and Peñas Coloradas Formations in the east, and no exposures of the Peñas Coloradas Formation have been observed west of the lahar (Abegglen, 2001; Tobler, 2001; Fig. 5). However, the thrust relationship, radiometrically dated strata, and transition in the heavy mineral assemblages obtained from the Peñas Coloradas and Chota Formations (Fig. 4; P-45, P-27) suggest that the Peñas Coloradas Formation is older than or at least coeval with the lower Chota Formation.

The Chota Formation was deposited in a fluvial to lacustrine environment. The dominance of magnesio-hastingite hornblende, basaltic brown hornblende, and clinopyroxene of diopsidic composition in the sandstones (Fig. 4) suggests a source region composed of andesitic and basaltic volcanic rocks. The abundance of folded volcanic sills and dykes in the Chota Formation suggests that part of the volcanic debris was derived from coeval volcanic centers. The presence of medium- to high-grade metamorphic minerals in the lower strata of the Chota Formation (P-27; Fig. 4) corroborates the stratigraphic succession on the basis of the radiometric evidence. A total of five ZFT and AFT ages from volcanic ash beds in the western sector of the Chota Formation range in age between 4.8±0.4 and 2.9±1.5 Ma (Table 1, Fig. 3), though the intense deformation renders it difficult to arrange the dated samples in correct stratigraphic order because of the low precision of some age data. An AFT age of 1.1±0.6 Ma (AT-13, Fig. 3),

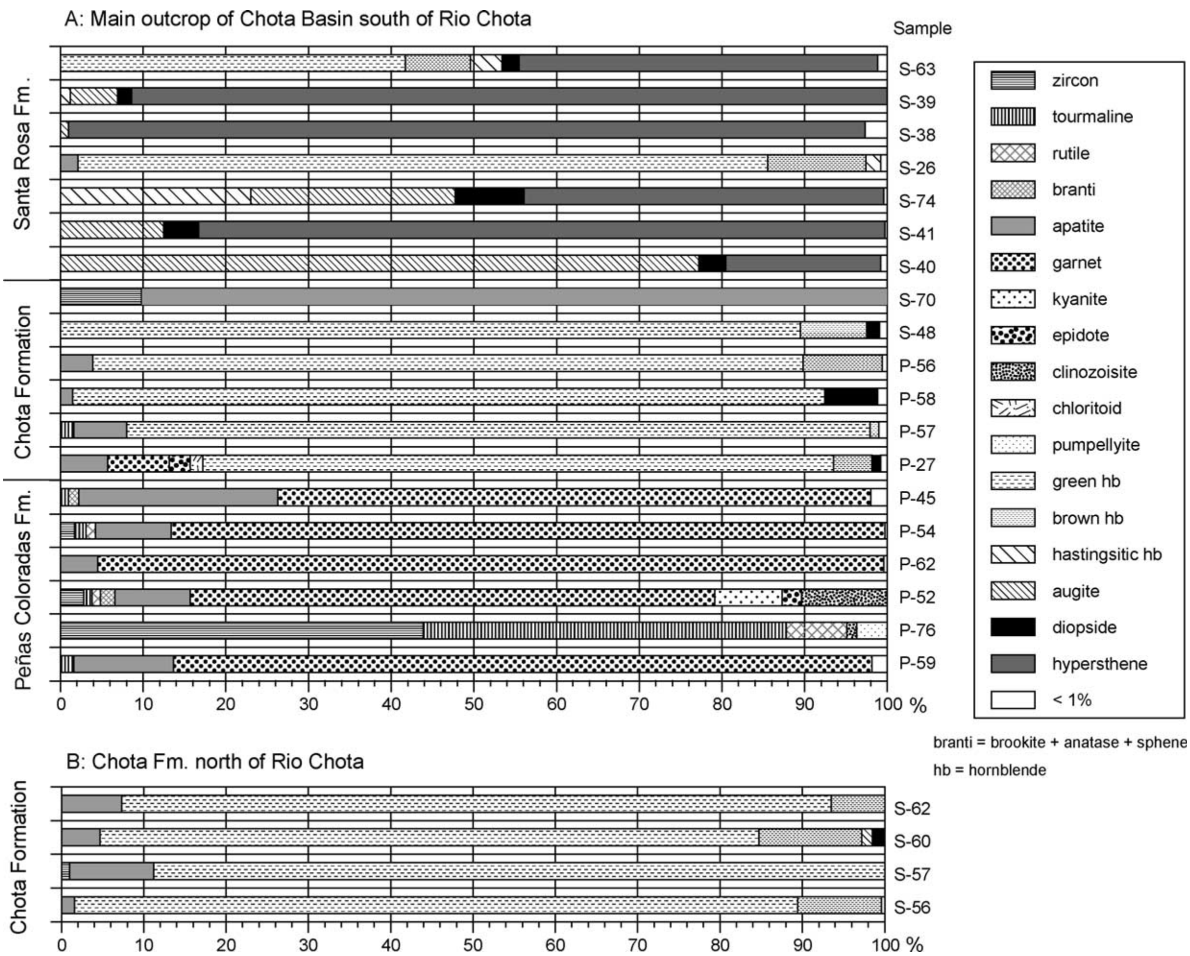

Fig. 4. Heavy mineral assemblages (frequency percentage) in sandstones from north and south of the Rio Chota in the Chota basin.

obtained from an ash bed 5–10 m below the lahar, is difficult to interpret, but we tentatively ascribe the ash to the Chota Formation (Abegglen, 2001).

The Santa Rosa Formation consists of fluvial to alluvial fan deposits, which thicken westward and prograde from the west over the Chota Formation (Baragán et al., 1996; Tobler, 2001). In addition to andesitic volcanic debris, the reworking of augite, hypersthene, and diopside suggests that the basic rocks of the Pallatanga unit in the CO contributed to the sedimentary flux (Fig. 4). The Gavilanes breccia represents a massive, 500 m thick debris flow horizon in the Santa Rosa Formation. The lack of internal layering suggests that it was deposited through a single catastrophic event, possibly due to basin margin collapse. The overlying upper section of the Santa Rosa Formation has an angular, unconformable contact (∼15°) with the Gavilanes breccia (Tobler, 2001).

The entire Chota basin series was deformed by postdepositional folding (Fig. 5). The dominant fold-and-thrust deformation phase occurred in an approximately WNW–ESE-oriented compressional field (e.g. Abegglen, 2001; Tobler, 2001). Deformed volcanic dykes and sills, frequently observed in the Chota Formation, predate the compressional deformation phase. Undeformed, tuffaceous, volcanoclastic rocks unconformably overlie the Chota Formation and yield a ZFT age of 0.5 ± 0.2 Ma (AT 6, Table 1, Fig. 3), which constrains the minimum age of the folding event. Baragán et al. (1996) infer a NW–SE-oriented, syndepositional extension during the deposition of the Chota Formation and a syndepositional, WNW–ESE-oriented compression during the deposition of the Peñas Coloradas and Santa Rosa Formations. Circumstantial arguments may imply that the Chota Formation was deposited in an approximately NW–SE-extending regime, which was preceded (during Peñas Coloradas deposition) and followed (during Santa Rosa deposition) by WNW–ESE-oriented compression. The conspicuous flip of sediment supply into the basin supports the Baragán et al. (1996) interpretation, though we propose a different stratigraphic model.

3.2. Quito–San Antonio–Guayllabamba basin

Sedimentary rocks of the Quito, San Antonio, and Guayllabamba basin unconformably overlie either basaltic rocks of the Cretaceous Pallatanga unit (CO) or Pliocene

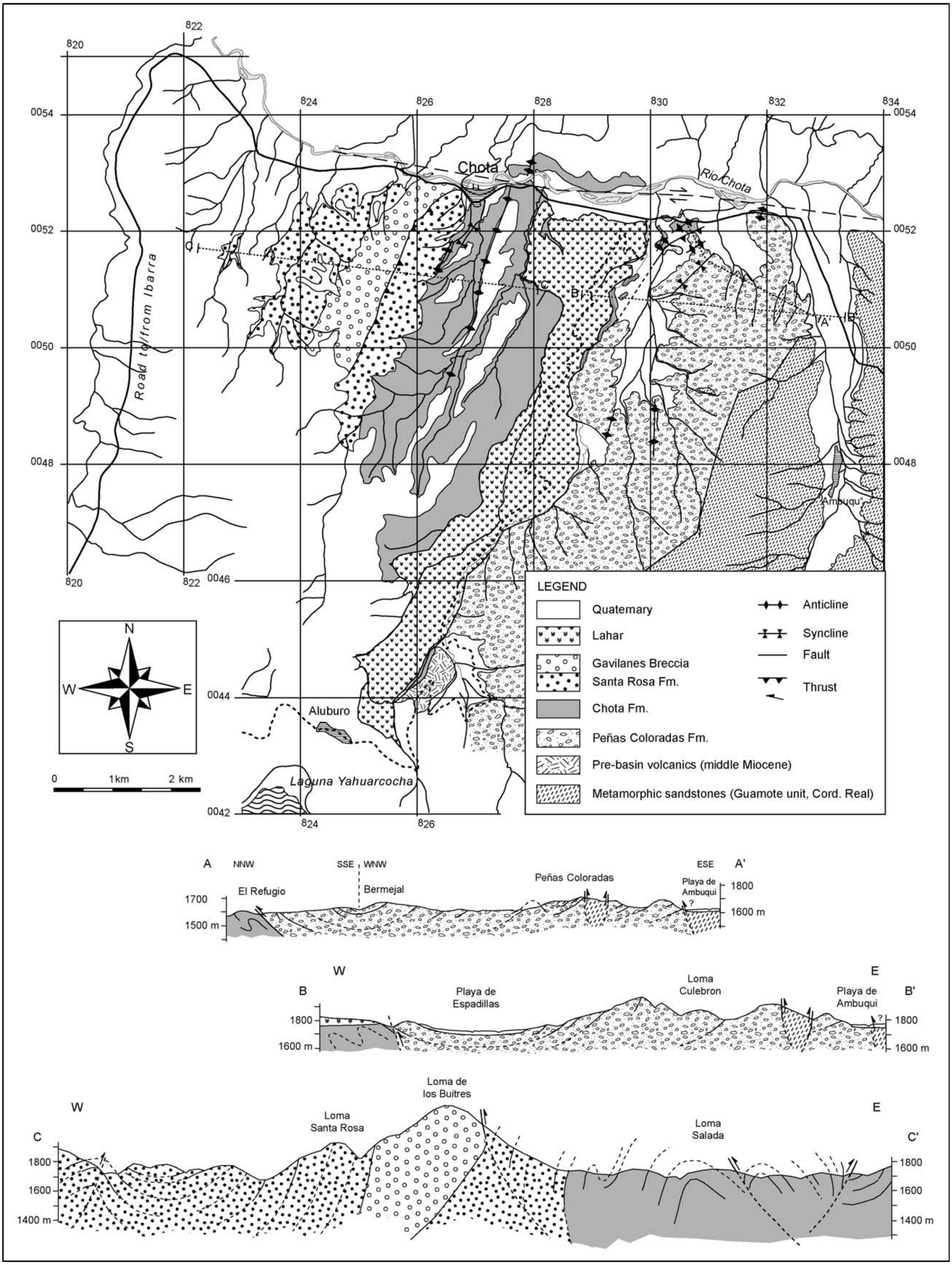

Fig. 5. Simplified geological map and cross-sections of the Chota basin (after Abegglen, 2001; Tobler, 2001). Note that the map and cross-sections are presented at different scales.

volcanic rocks of the basal Pisque Formation (Villagómez et al., 2002). The younger sedimentary sequence consists of a complex stack of volcanic and volcanoclastic deposits (Alvarado, 1996; Ego and Sébrier, 1996; Lavenu et al., 1996; Villagómez et al., 2002). Radiometric ages of strata are rare, though correlations between volcanic edifices and the strata (e.g. OLADE-INECEL, 1980; Barberi et al., 1988; Geotermia Italiana, 1989; Fig. 2) suggest the basin sequence is 6–5 Ma old or younger. $^{40}Ar/^{39}Ar$ radiometric analyses of ferromagnesian minerals currently are in progress.

The basal Pisque Formation (Fig. 2), which includes the basal Lava member, consists of andesitic and basaltic lavas unconformably overlain by tuffs and lahars of the lower Pisque Formation (Golden Tuffs and Puente Viejo members; Villagómez, 2003). The basal Lava member has been correlated with the Casitagua, Culbiche, and Chicaloma volcanoes, which yield K/Ar whole-rock ages of 2.25 ± 0.25, 2.6 ± 0.06, and 3.46 ± 0.1 Ma (OLADE-INECEL, 1980; Barberi et al., 1988), which places the basal Pisque Formation in the late Pliocene. Fluvial and alluvial fan facies prevail in the upper Pisque Formation. The San Miguel Formation is a volcanic, tuff-rich sequence of eastward prograding deltaic and lacustrine deposits, the latter of which were strongly deformed during synsedimentary, gravity-driven displacements in response to the loading exerted by younger lahars deposited during the deposition of the Guayllabamba Formation (Villagómez, 2003). The overlying Guayllabamba Formation records a period of intense volcanism and compressive tectonic activity. To the west and south, primary volcanic deposition (lavas, pyroclastic flows, avalanches) prevailed, whereas in the east, lahar flows filled a former lake that existed during the San Miguel era. Several volcanic domes also intruded the lake deposits. Radiometric ages of the volcanic deposits, which have been correlated with the Guayllabamba Formation (K/Ar andesite, 1.62 ± 0.16 Ma, OLADE-INECEL, 1980; K/Ar rhyolite, 0.98 ± 0.13 Ma, Barberi et al., 1988), indicate Pleistocene extrusion ages. Alluvial fan-related conglomerates cap the Guayllabamba Formation and can be correlated with the growth of the Calderón–Catequilla fold structure (Villagómez, 2003). In addition, the east-verging, reverse Quito fault system became active during the final stages of the deposition of the Guayllabamba Formation and separated the San Antonio from the Guayllabamba basin.

The Chiche Formation (Fig. 2) was deposited in calm, low energy, lacustrine, and fluvial environments in the Quito and Guayllabamba basins, though several lahars flowed into the depocenters. The presence of the fossil teeth *Glossotherium* (Lavenu et al., 1996) at the top of the Chiche Formation is commonly interpreted to indicate a middle Pleistocene age (≈ 0.5 Ma). Compressive deformation was revived in the area during the waning stages of deposition of the Chiche Formation, when the southern segments of the Quito fault system became active (Quito and Botadero faults). The overlying Machángara Formation in the Quito area, the lower part of which is composed of primary volcanic rocks and the upper part of epiclastic deposits, shows progressive unconformity geometries with the Quito fault (Villagómez, 2003). The coeval Mojanda Formation to the north was derived from the middle-late Pleistocene Mojanda volcanic complex (0.6 Ma, K/Ar whole-rock andesite, Barberi et al., 1988). This stratigraphic relationship suggests that the Quito fault has been active until recently. Widespread, unconsolidated, airfall tuffs, with distinct pedogenetic intervals, constitute the Cangahua Formation, which disconformably overlies older formations. Radiocarbon, obsidian-hydration, and thermoluminescence data suggest that the Cangahua Formation probably accumulated during the past 100 Ka (Clapperton and Vera, 1986).

Previous studies in the Quito–San Antonio–Guayllabamba basin have proposed that it formed in either an E–W- or NNE–SSW-trending extensional regime (e.g. Tibaldi and Ferrari, 1992; Ego and Sébrier, 1996). However, the timing of normal fault activity is stratigraphically poorly constrained. Recent stratigraphic and structural observations (Villagómez, 2003) suggest that approximately E–W-oriented extension occurred during the Pliocene, as documented by the approximately N–S-striking, mesoscale (metric), synsedimentary normal faults, which only displace the Pisque and San Miguel Formations. Limited age determinations and field mapping (Villagómez, 2003) suggest that E–W-oriented compression commenced during the deposition of the Guayllabamba Formation (~ 1 Ma) and initiated the formation of the Calderón-Catequilla fold. The prevailing stress field was presumably similar to that observed today, as determined from shallow focal earthquake solutions in the region (mean orientation N98°E; Ego and Sébrier, 1996). Tectonic compression probably decreased during the deposition of the Chiche Formation, though it recommenced from ~ 0.5 Ma to the present and is responsible for neotectonic activity along the Quito and Botadero reverse faults (Villagómez, 2003).

3.3. Ambato–Latacunga basin

Lavenu et al. (1992) recognize four Pliocene—Recent formations stratigraphically above volcanic rocks in the Ambato–Latacunga basin. The basin strata wedge out toward the CO in the west (Fig. 2), which documents the creation of the positive relief along the western margin of the basin. The local nomenclature of the underlying volcanic formations is somewhat enigmatic. However, chronostratigraphic data and lithologic descriptions (Baldock, 1982; Egüez et al., 1992; Lavenu et al., 1992, 1995, 1996; Ego and Sébrier, 1996) imply that these formations correspond to Oligocene-Miocene volcanic rocks, which are widely distributed in the southern Ecuadorian arc and forearc (e.g. British Geological Survey-GODIGEM, 1997; Hungerbühler et al., 2002), where they are regionally referred to as the Saraguro group. The local Pisayambo and

Alausí Formations (Fig. 2) are time equivalents of the Turi and Tarqui Formations in the Cuenca area to the south (Steinmann et al., 1999).

Volcanic and volcanoclastic (fluvial and lacustrine) rocks of the Sicalpa Formation overlie volcanic rocks of the Turi and Tarqui Formations in the Latacunga area. Acidic tuffs intercalated within the Sicalpa yield late Miocene K/Ar radiometric ages of 3.59 ± 0.28 and 2.65 ± 0.21 Ma (Lavenu et al., 1992). The overlying lower Latacunga Formation mainly has a volcanic origin and is composed of lahars, lava flows, volcanic breccias, and fluvial deposits at the top. The upper Latacunga Formation consists of lacustrine and fluvial deposits (Lavenu et al., 1995). Andesite whole-rock and plagioclase K/Ar analyses close to the boundary between the lower and upper Latacunga Formation (Lavenu et al., 1992, 1995) yield late Pliocene-early Pleistocene ages (1.85 ± 0.19, 1.73 ± 0.35, 1.4 ± 0.29 Ma, Fig. 2). The unconsolidated pyroclastic Chalupas Formation unconformably overlies the Latacunga Formation. An andesitic horizon in the Chalupas Formation yields a whole-rock K/Ar age of 1.21 ± 0.05 Ma (Barberi et al., 1988).

The Ambato–Latacunga basin is bound by thrusts in the east (east-dipping Pisayambo fault) and west (west-dipping La Victoria fault). Both faults represent a segment or branch of the Peltetec and Pallatanga–Pujilí–Calacalí fault systems, respectively. Stratigraphic evidence and synsedimentary folding revealed by progressive unconformity development in the Latacunga Formation suggest that a significant phase of compressive deformation occurred between ~ 1.85 and ~ 1.2 Ma (Lavenu et al., 1995, 1996).

3.4. Alausí–Riobamba basin

Sedimentary rocks of the Alausí–Riobamba basin unconformably overlie basement rocks of the Pallatanga unit and a thick sequence of Oligocene-Miocene volcanic rocks, which can be distinguished into two formations. Volcanic facies observations and radiometric age determinations from the lower Huigra Formation (Egüez et al., 1992; Fig. 2) suggest that it correlates with the Saraguro group. Similarly, radiometric ages (8.12 ± 0.10, 7.10 ± 0.03 Ma, K/Ar andesite whole-rock; Barberi et al., 1988; Lavenu et al., 1996) from the Upper Alausí Formation (Fig. 2) correlate these volcanic rocks with the late Miocene volcanoclastic and volcanic Turi and Tarqui Formations (e.g. Hungerbühler et al., 2002) in the southern Ecuadorian forearc. Although no physical contact was observed, the lacustrine Sicalpa Formation is assumed to overlie the Alausí volcanics unconformably (Lavenu et al., 1992). The undated, thick, alluvial fan and fluvial conglomerates of the Palmira Formation overlie the Pliocene Sicalpa Formation with an angular unconformity (Egüez et al., 1992; Lavenu et al., 1996; Fig. 2). The presence of these coarse sediments, which were derived from the east, indicates a significant tectonic and/or climatic change during the late Pliocene (Lavenu et al., 1996). These coarse facies may have been deposited during synsedimentary deformation in the contiguous Latacunga Formation, and a compressive tectonic regime may have prevailed during the deposition of the Palmira Formation.

4. Discussion

The Ecuadorian Andes lie in the southern part of the northern Andean block (Fig. 6), which is displacing to the NNE relative to the Guyana shield in response to the eastward subduction of the Nazca plate and the northwestward drift of the South American plate. Displacement of the block is accompanied by approximately E-W-oriented shortening, overthrusting onto the Caribbean plate (Pennington, 1981; Kellogg and Bonini, 1982; Ego et al., 1996), and right-lateral displacement, which is accommodated in Ecuador and Colombia by regional-scale, strike-slip faults (Toussaint and Restrepo, 1994). The inception of and subsequent subsidence in the Gulf of Guayaquil and neighboring forearc areas to the east (Benítez, 1995; Hungerbühler et al., 2002; Witt, 2002) was a result of the displacement of the northern Andean block after 16–15 Ma. Geometric and stratigraphic relationships suggest the southern tip of the northern Andean block has displaced approximately 100 km northward (Shepherd and Moberly, 1981; Hungerbühler, 1997). To link the Pallatanga fault in the Gulf of Guayaquil kinematically with the main eastern boundary of the northern Andean block in Colombia and Venezuela (eastern Andean front fault zone; Toussaint and Restrepo, 1994; Ego et al., 1996), a restraining bend across the northern Ecuadorian Andean chain must be inferred. This tectonic structure has not been identified, though it probably includes fragments of the Pallatanga–Pujilí–Calacalí, Peltetec, and Chingual-La Sofía faults, as well as parts of the frontal fault zone of the Andes in southern Colombia (e.g. Ego et al., 1996; Fig. 6).

Late Miocene (6–4 Ma) compressive tectonic activity drove high exhumation rates in the Andean chain near the Chingual-La Sofia fault along the Ecuadorian–Colombian border (Spikings et al., 2000) (Figs. 1 and 6b). Neotectonic activity of the fault has displaced latest Pleistocene–Holocene volcanic flows and postglacial river valleys in the tributaries of the El Dorado (Ferrari and Tibaldi, 1992). Furthermore, AFT data from a mylonitic shear band in the Chimbo-Toachi shear zone (Fig. 1) suggest high cooling and exhumation rates in the Macuchi unit since 5 Ma (Spikings et al., 2005). Similarly, AFT data from the sub-Andean thrust belt (Fig. 1), located between the Cosanga and sub-Andean faults, records rapid cooling and exhumation between 6 and 2.5 Ma (Ruiz, 2002). These data correlate with higher cooling rates and increased depths of exhumation in the northern CR (north of 1°30′S) during the past 6 Ma, relative to the southern CR (Spikings et al., 2001). Collectively, this information suggests that the restraining

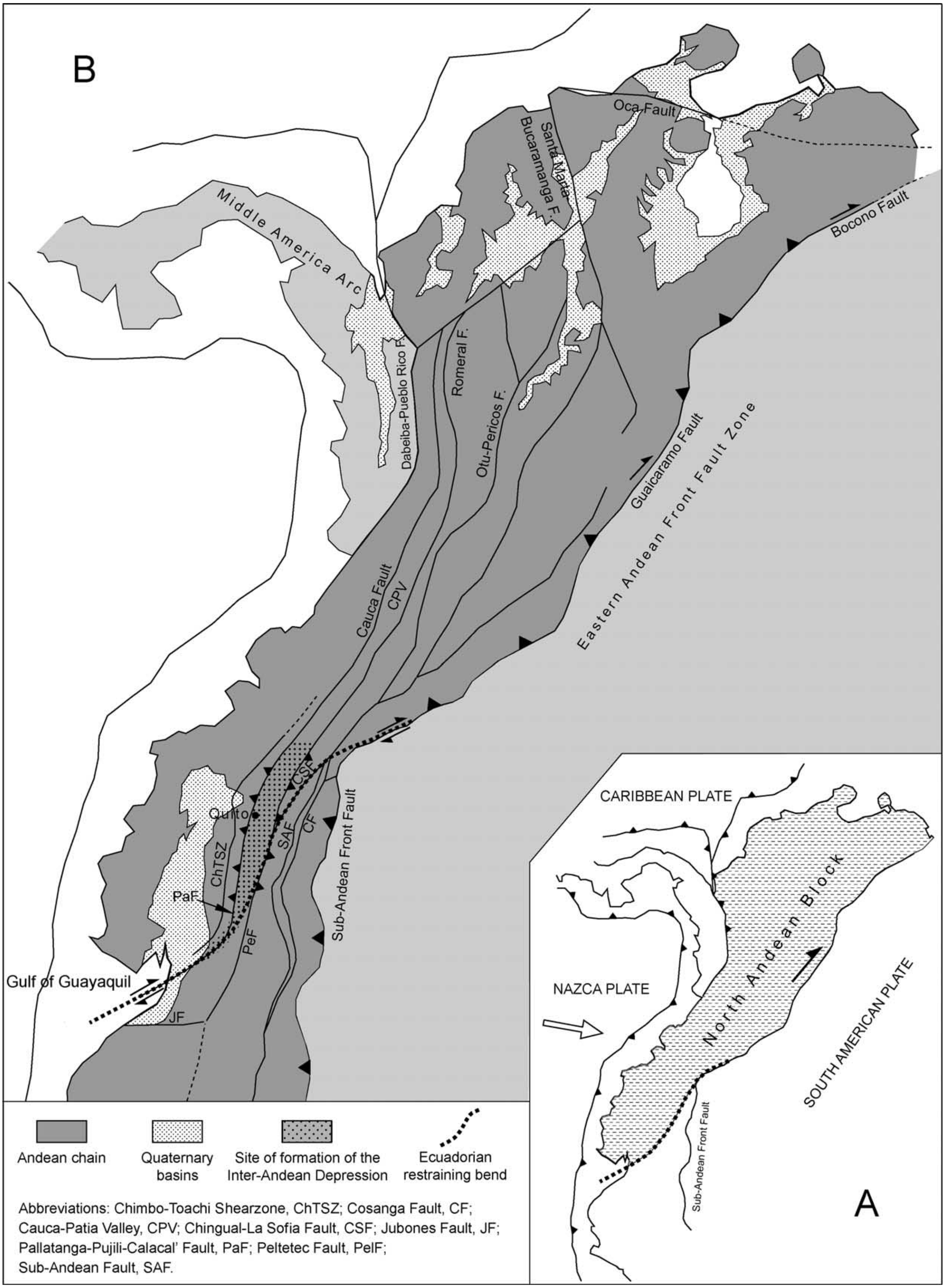

Fig. 6. Schematic location of the north Andean block (A) and simplified map of the northern Andes showing the main faults and tectonic boundaries (B). Note that the tectonic regime is transpressive due to the subduction of the Nazca plate. The southern tip of the north Andean block is bound to the east by the Ecuadorian restraining bend, which represents a composite structure and includes segments of the Pallatanga-Pujilí-Calacalí, Peltetec, and Chingual-La Sofía faults. Modified from Pennington (1981), Kellogg and Bonini (1982), Toussaint and Restrepo (1994) and Ego et al. (1996).

bend has been active since at least 15 Ma and that a significant pulse of activity commenced at 6 Ma.

According to our results, it is plausible that the IAD structure formed as a result of displacement along the restraining bend since 6 Ma (Fig. 6). Several late Miocene–Pleistocene (?) subbasins are preserved within the elongated IAD. Sedimentation in the subbasins commenced during the latest Miocene (6–5 Ma) in the north (Chota basin) and spread to the southern basins (Fig. 2) during the Pliocene (4–3 Ma).

The E–W-oriented, synsedimentary, compressional deformation in the Ambato–Latacunga basin is manifested by low amplitude folding with internal, progressive unconformities during 1.85–1.2 Ma (Lavenu et al., 1995). The deformation was driven by opposite-verging reverse faulting along the basin margins (La Victoria and Pisayambo faults; Fig. 1), which resulted in horizontal shortening across the IAD at an estimated rate of approximately 1.4 ± 0.3 mm/year (Lavenu et al., 1995).

Sedimentary sequences in the Quito–San Antonio–Guayllabamba basin were deformed by the N–S-trending, east-verging Calderon–Catequilla and Quito folds and the Quito and Botatero thrust faults (Ego and Sébrier, 1996; Villagómez, 2003). Tentative estimates suggest that the main phase of compressive tectonic activity started during the deposition of the Guayllabamba Formation at approximately 1.0 Ma (Ego and Sébrier, 1996; Villagómez, 2003). However, N–S-trending synsedimentary normal faults in the Pisque Formation indicate E–W-oriented extension (Villagómez, 2003), which conflicts with Ego and Sébrier's (1996) statistical reconstruction of a N–S-trending extensional regime that would have disrupted the Pisque Formation prior to folding.

The WNW–ESE-oriented compression has generated NNE–SSW-trending fold axes and thrust planes in the Chota basin (Fig. 5). The folding is sealed by 0.5 My old, approximately horizontal, bedded volcanoclastic rocks (Tobler, 2001; Winkler et al., 2002), which constrain the minimum age of basinwide deformation in the Chota basin. Baragán et al. (1996) propose a structural reconstruction of the Chota basin sequence using the microtectonic data inversion method and differentiate between synsedimentary and postsedimentary deformation. In translating their results to the revised chronostratigraphic framework presented here (e.g. the Peñas Coloradas Formation is considered the oldest formation in the basin), NW–SE to WNW–ESE compressive stress prevailed throughout the entire basin history. Synsedimentary, small-heave (cm to m) normal faults were detected in local parts of the fine-grained, thinly bedded Chota Formation, which indicates that extension was oriented between WNW–WSE and WSW–ENE (Abegglen, 2001; Tobler, 2001). These results are similar to those of Baragán et al. (1996), who observed NW–SE syndepositional extension in the Chota Formation, and consistent with the W–E-oriented extensional stress field inferred for the coeval Pisque Formation in the Quito–San Antonio–Guayllabamba basin (Villagómez, 2003). In the Chota series, the distinct change in source material—from a granitic/metamorphic terrane, which shed into the Peñas Coloradas Formation in the east, to a volcanic and mafic terrane, which supplied the Santa Rosa Formation in the west—is consistent with an initial phase of compression-related rock uplift and exhumation in the bordering CR, followed by uplift in the CO. The massive Gavilanes breccia in the Santa Rosa Formation may represent a huge structural failure driven by tectonic oversteepening of the slopes along the western basin margin. The Chota Formation occupies a transitional position with respect to stratigraphy and probably was deposited between the prominent alluvial fans, which shed into the basin. The fluvial to lacustrine deposits may document a period of decreased compressive tectonic activity or even minor extension, which resulted in volcanic dyke and sill emplacement.

Various models have defined a single basin type for the IAD and interpreted the origin of the IAD within that context. There is general agreement that a transpressive tectonic regime prevailed during most of the history of the IAD structure (e.g. Ego et al., 1996; Ego and Sébrier, 1996). Alternative arguments that it may have formed as an extensional graben or in a transtensional setting can be disproven. The IAD formed during a period of high rates of exhumation throughout the cordilleras of northern Ecuador (Spikings et al., 2000, 2001, 2005), and it is unlikely that deep-seated subsidence would occur in the crest of the exhuming fault blocks. Regardless of the considerable northward displacement of the north Andean block ($\sim$100 km; Shepherd and Moberly, 1981), this distance is not sufficient because of the nature of strain partitioning in transcurrent systems needed to generate pull-apart basin subsidence on the scale of the IAD. However, the presence and provenance of the clastic material in the IAD clearly shows that differential subsidence of the IAD occurred with respect to both the bordering cordilleras. The IAD may have developed as a ramp valley (Fig. 7; Willis, 1928; Cobbold

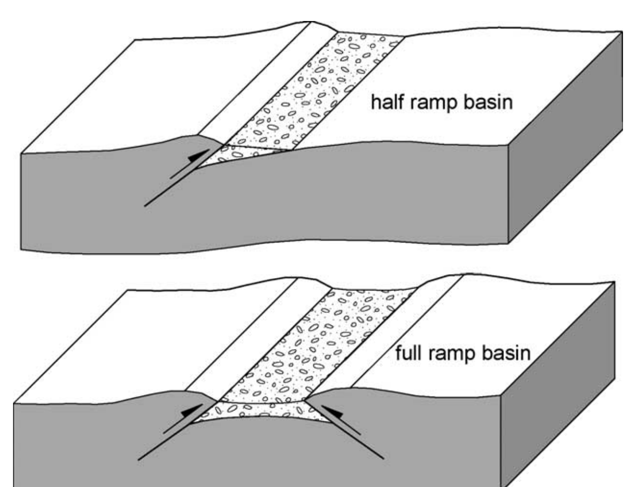

Fig. 7. Schematic presentation of the full- and half-ramp basin models. After Cobbold et al. (1993).

et al., 1993), in that fault blocks of the bordering cordilleras may have overthrust the basin floor. Mann et al. (1991) account for Miocene-Pliocene basin structures in Hispañola by applying a similar tectonic evolution along the restraining bend between the North American and Caribbean plates.

Lavenu et al. (1996) propose that a compressional (push down) setting prevailed in the Ambato-Latacunga area, which corroborates a full-ramp setting (Cobbold et al., 1993). These authors assume that both the east-verging Victoria fault in the west and the west-verging Pisayambo fault in the east (Fig. 1) overthrust the basin margins. In the almost flat basin floor, several low amplitude anticlines and synclines, with approximately basin-parallel trends, developed in the syntectonic basin fill series and underlying volcanic rocks. The syndepositional nature of the tectonic activity is documented by progressive unconformities. A model calculation estimates that the total shortening between the basin margins, which are currently 25 km apart, is 3400 ± 800 m (Lavenu et al., 1995).

No sedimentary rocks have been identified from the eastern margin (CR) of the Quito–San Antonio–Guayllabamba basin (Villagómez, 2003), which suggests that active reverse faulting was absent along this margin during the formation of the basin. However, the accumulation of 1200–1400 m of sedimentary and volcanic rocks implies that a topographic barrier existed. The major basin-bounding fault was the Pallatanga–Pujilí–Calacalí to the west (Fig. 1), and shortening along the western basin margin, which resulted in the Quito and Botadero faults and the Calderon–Catequilla fold, implies that the Pallatanga–Pujilí–Calacalí fault had a significant reverse component. This sector of the IAD therefore may be referred to as a half-ramp basin (Fig. 7; Cobbold et al., 1993), though the paucity of field observations prevents us from denying the possibility that an opposing reverse fault in the east may have been active.

The amount of shortening in the Chota basin is probably the greatest within the entire Ecuadorian IAD (Fig. 4). A predominantly compressive regime has been confirmed by structural analyses (Baragán et al., 1996), though no attempt has been made to quantify the shortening. The presence of reverse faults on both basin margins (Pallatanga–Pujilí–Calacalí and Peltetec faults; Fig. 1) can be inferred only from circumstantial evidence provided by the radiometrically established stratigraphic succession, sedimentary facies, and the provenance of detrital grains (Figs. 2–4). Reverse faulting probably commenced in the east and exposed metamorphic rocks of the CR, which eroded to produce massive alluvial fan systems (Peñas Coloradas Formation) that drained to the west. At a stratigraphically poorly defined time, the western basin margin became the major source of debris, shedding alluvial fans and catastrophic flows into the basin (Santa Rosa Formation, including the Gavilanes breccia). The basic detrital grains observed in the Santa Rosa Formation suggest that the CO was exhuming, probably via reverse activity along the Pallatanga-Pujilí-Calacalí fault. The transition of reverse faulting from the eastern margin to the western margin is recorded by the fluvial and lacustrine Chota Formation, which implies a reduction in the regional, net compressive stress field. Small-scale, synsedimentary, normal faulting in the lacustrine Chota Formation may be interpreted as gravity-driven tension during the transition of reverse faulting from the eastern to the western basin margin.

5. Conclusions

The elongate IAD structure formed in a right-lateral transpressive tectonic regime during a period of high rates of exhumation throughout the cordilleras in northern Ecuador after 6–5 Ma. During its development, several subbasins formed in which alluvial fan, fluvial, lacustrine facies, and contemporaneous volcanic products were deposited. The chronostratigraphic framework for the sedimentary series of the Chota basin, combined with field observations and data from the Quito–San Antonio–Guayllabamba, Ambato–Latacunga, and Alausí–Riobamba basins, implies that sedimentation spread from the north to the south after 6–5 Ma. Synsedimentary compressive deformation is documented by reverse fault movements along the basin margins and folding and faulting within the basins. A full-ramp (locally half-ramp) basin model (Fig. 7), in which opposite verging reverse faults drive differential uplift of the basin-bordering cordilleras with respect to the IAD, most appropriately describes the tectonosedimentary assemblages. Minor synsedimentary normal faulting may be attributed to gravity-driven extension in the oversteepened orogenic wedge during periods of waning compression. The north-to-south progradation of compression and basin formation possibly correlates with a southward shift of the subducting, buoyant Carnegie ridge (e.g. Spikings et al., 2001). The IAD ramp basin formed in the vicinity of a major restraining bend, which accommodates general northward displacement of the north Andean block (Fig. 6).

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